

Personal data

Name: Mohamed Elobaid

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Research fields

Optimization-based design and control of robotic systems with guarantees

Current position

Research Scientist KAUST, since Sept. 2025.

Previous positions

Postdoctoral Researcher, Istituto Italiano di Tecnologia. Responsible for the joint research initiative with Honda R&D on humanoid robot avatars, and collaborator on the design and control of ergoCub new generation of humanoids, Feb. 2022 – Aug. 2025.

Graduate Teaching Assistant, Sapienza University of Rome, Jan. 2021 – Oct. 2021.

Robotics Engineer Fellow, Istituto Italiano di Tecnologia. Main responsible for the iCub loco-manipulation teleoperation project, Jan. 2018 – Nov. 2018.

Automation Engineer, Kenana Engineering and Technical Services. Engineering and commissioning on several new and upgraded sugar factories automation systems, Jan. 2012 – May 2015.

Teaching Assistant, University of Khartoum, Aug. 2011 – Aug. 2012.

Education and titles

Ph.D. in Automatic Control, Sapienza University of Rome in *cotutelle* with Université Paris-Saclay, *magna cum laude*. Advisors: Prof. Salvatore Monaco and Dr. Dorothée Normand-Cyrot, Nov. 2018 – Feb. 2022.

M.Sc. in Control Engineering, Sapienza University of Rome, 110/110 *cum laude*, Sept. 2015 – Oct. 2017.

B.Sc. (Hons) in Electrical and Electronic Engineering, University of Khartoum, second class upper division, Nov. 2006 – Aug. 2011.

Awards and individual grants

2024 Best Paper Award, IEEE-RAS Technical Committee on Whole-Body Control, for the preceding year.

2023 Finalist, IEEE ICRA Outstanding Paper Award on Human–Robot Interaction.

2019 Best Paper finalist, Intelligent Systems Conference IntelliSys.

2019 Programme Vinci grant, Université Franco Italienne, grant no. cap. 2 Vinci 2019 C2-897.

International journal papers

Published or in press

- J7 C. Sartore, **M. Elobaid**, L. Rapetti, G. Romualdi, S. Daffarra, N. A. Piga, I. Sorrentino, P. M. Viceconte, S. Traversaro, U. Pattacini, L. Fiorio, F. Draicchio, G. Tranfo, L. Natale, M. Maggiali and D. Pucci, “Towards shared embodied intelligence in humanoid robots through optimization, development and testing of the human-aware ergoCub robot,” *Nature Machine Intelligence*, vol. 8, pp. 1221–1237, 2026.
- J6 D. Gorbani, **M. Elobaid**, G. L’Erario, H. A. O. Mohamed and D. Pucci, “Data-fused MPC with guarantees: application to flying humanoid robots,” *IEEE Control Systems Letters*, vol. 9, pp. 2909–2914, 2025.
- J5 **M. Elobaid**, G. Turrise, L. Rapetti, G. Romualdi, S. Daffarra, T. Kawakami, T. Chaki, T. Yoshiike, C. Semini and D. Pucci, “Adaptive non-linear centroidal MPC with stability guarantees for robust locomotion of legged robots,” *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2806–2813, 2025.
- J4 C. Cardenas-Perez, G. Romualdi, **M. Elobaid**, S. Daffarra, G. L’Erario, S. Traversaro, P. Morerio, A. Del Bue and D. Pucci, “XBG: end-to-end imitation learning for autonomous behaviour in human–robot interaction and collaboration,” *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11617–11624, 2024.
- J3 S. Daffarra, U. Pattacini, G. Romualdi, L. Rapetti, R. Grieco, K. Darvish, G. Milani, E. Valli, I. Sorrentino, P. M. Viceconte, A. Scalzo, S. Traversaro, C. Sartore, **M. Elobaid**, N. Guedelha, C. Herron, A. Leonessa, F. Draicchio, G. Metta, M. Maggiali and D. Pucci, “iCub3 avatar system: enabling remote fully immersive embodiment of humanoid robots,” *Science Robotics*, vol. 9, no. 86, eadh3834, 2024.
- J2 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Station-keeping of L_2 halo orbits under sampled-data model predictive control,” *Journal of Guidance, Control, and Dynamics*, vol. 45, no. 7, pp. 1337–1346, 2022.
- J1 **M. Elobaid**, S. Monaco and D. Normand-Cyrot, “Approximate transverse feedback linearization under digital control,” *IEEE Control Systems Letters*, vol. 6, pp. 13–18, 2020.

Submitted papers

- J8 **M. Elobaid** and E. Feron, “Antipodal initialization for box-constrained quadratic programs,” submitted to *IEEE Transactions on Automatic Control*, April 2026, under review.

Proceedings of refereed international conferences

- C11 **M. Elobaid**, S. Park and E. Feron, “On the stabilization of rigid formations on regular curves,” IFAC World Congress, 2026, to appear.
- C10 S. Taliani, G. Nava, G. L’Erario, **M. Elobaid**, G. Romualdi and D. Pucci, “Online nonlinear MPC for multimodal locomotion,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA 2025)*, pp. 12344–12351, 2025.
- C9 **M. Elobaid**, S. Daffarra, E. Ranjbari, G. Romualdi, T. Chaki, T. Kawakami, T. Yoshiike and D. Pucci, “Remote telepresence over large distances via robot avatars: case studies,” in *Proc. IEEE International Conference on Telepresence*, 2024. [TODO: page numbers.]
- C8 L. Rapetti, C. Sartore, **M. Elobaid**, Y. Tirupachuri, F. Draicchio, T. Kawakami, T. Yoshiike and D. Pucci, “A control approach for human–robot ergonomic payload lifting,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA 2023)*, pp.

7504–7510, 2023. *ICRA Outstanding HRI Paper finalist; IEEE-RAS TC-WBC Best Paper Award 2023.*

- C7 **M. Elobaid**, G. Romualdi, G. Nava, L. Rapetti, H. A. O. Mohamed and D. Pucci, “Online non-linear centroidal MPC for humanoid robot payload carrying with contact-stable force parametrization,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA 2023)*, pp. 12233–12239, 2023.
- C6 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Virtual holonomic constraints for Euler–Lagrange systems under sampling,” in *Proc. European Control Conference (ECC 2022)*, pp. 693–698, 2022.
- C5 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Digital path-following for a car-like robot,” IFAC Control Conference Africa, 2021, *IFAC-PapersOnLine*, vol. 54, no. 21, pp. 174–179.
- C4 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “On stable right-inversion of non-minimum-phase systems,” in *Proc. IEEE Conference on Decision and Control (CDC 2020)*, pp. 5153–5158, 2020.
- C3 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “Sampled-data tracking under model predictive control and multi-rate planning,” IFAC World Congress, 2020, *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 3620–3625.
- C2 **M. Elobaid**, M. Mattioni, D. Normand-Cyrot and S. Monaco, “On unconstrained MPC through multirate sampling,” IFAC Symposium on Nonlinear Control Systems, 2019, *IFAC-PapersOnLine*, vol. 52, no. 16, pp. 388–393.
- C1 **M. Elobaid**, Y. Hu, G. Romualdi, S. Daffarra, J. Babič and D. Pucci, “Telexistence and teleoperation for walking humanoid robots,” in *Proc. Intelligent Systems Conference (IntelliSys 2019)*, pp. 1106–1121, 2019. *Best Paper finalist.*

Submitted papers

- C13 A. Alshedayfat, A. Abualsaud, A. Al Nasser, O. Wali, A. Laing-Herbert, S. M. Alam, M. O. Tahir, **M. Elobaid** and E. Feron, “Design, control and field validation of a USV–TUV system,” submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).
- C12 Y. Toyomoto, **M. Elobaid**, B. Ferguson, A. Quattrini Li and S. Park, “Safe ergodic control for multi-robot systems via quadratic programming,” submitted to the IEEE International Conference on Robotics and Automation (ICRA 2027).

Ph.D. thesis

A sampled-data approach in control problems involving partial dynamics cancellation, advisors Prof. Salvatore Monaco and Dr. Dorothée Normand-Cyrot, Sapienza University of Rome and Université Paris-Saclay, 2022.

Invited talks and workshops organization

- W1 *Safety-Critical Software and Control for Robotics* workshop co-organization as part of IEEE ARM 2026, Evry, Paris, July 2026.
- P6 *Robust locomotion via online nonlinear MPC*, Honda R&D Engineering, Saitama, Japan, 2023.
- P5 *Feedback linearization via digital control*, tutorial series for researchers at the Istituto Italiano di Tecnologia, Genoa, Italy, 2022.

- P4 *Path-following under digital control*, annual meeting of the Società Italiana Docenti e Ricercatori in Automatica, Ancona, Italy, 2021.
- P3 *Transverse feedback linearization under digital control*, American Control Conference, 2021.
- P2 *On stable right-inversion of partially minimum-phase systems*, IEEE Conference on Decision and Control, 2020.
- P1 *On unconstrained MPC under sampling*, 11th IFAC Symposium on Nonlinear Control Systems, Vienna, Austria, 2019; repeated at the SIDRA annual meeting, 2020.

Funded research projects

Sept. 2026 - Aug. 2027 OptiStep: hardware-and-control co-design of an adaptive active ankle-knee prosthesis. KAUST-IIT Joint Call seed project. Proposal conceived and written by M. Elobaid; scientific lead for the KAUST side. Concurrent optimization of hardware parameters and control strategy for a powered, terrain-adaptive lower limb prosthesis.

Feb. 2022 – Jan. 2025 Honda-IIT joint research on robot avatars. Industrial research project on teleoperation and control of humanoid robots for grasping, positioning and placing packages in warehouse-like environments. Technical lead for IIT.

Jan. 2024 – June 2024 Software suite for the teleoperation and control of Workfar's Syntro. Technology transfer project delivering a full-stack teleoperation and control software for the Syntro robot. Role: Technical Lead and Project Responsible.

Further professional and scientific activities

Member of the IEEE Control Systems Society and the IEEE Robotics and Automation Society. Reviewer, over the past seven years, for IEEE Transactions on Robotics, IEEE Robotics and Automation Letters, IEEE Control Systems Letters, and for ACC, CDC, ECC, ICRA, IROS, Humanoids and the IFAC World Congress.

Invited committee member, IFAC Control Conference Africa. 2024.

Teaching experience

Jan. 2021 – Oct. 2021 Graduate Teaching Assistant, Sapienza University of Rome. Examination organization for *Nonlinear Systems and Control* (M.Sc.) and *Teoria dei Sistemi* (B.Sc.).

Aug. 2011 – Aug. 2012 Teaching Assistant, University of Khartoum. Tutorials for *SC5121 Linear Control Theory* and *SC5226 Design of Control Systems*; instructor for *SC4210/SC5124/SC5228 Control Systems Lab I-III*, *EC1102 Computer Systems Lab I*.

2012, 2015, 2017, 2022 Outreach workshop *Control engineering practice and prospects*, for final-year control and instrumentation students, University of Khartoum.

Non-thesis Research supervision

Present

Mohannad Alhakami, KAUST, Bilevel Optimization based co-design of Prosthesis, since Sept. 2026.

Altayebmohd Alshedayfat, KAUST, Marine robotics; first results in [C13], since Oct. 2025.

Ali Abualsaud, KAUST, Marine robotics; first results in [C13], since Oct. 2025.

Yo Toyomoto, KAUST, Ergodic Control, first results in [C12], since Feb. 2026.

Past

Davide Gorbani, Istituto Italiano di Tecnologia, supervised his work on data-driven model predictive control in [J6], July 2025 - Dec. 2025.

Carlos Cardenas-Perez, Istituto Italiano di Tecnologia, co-supervised his work on end-to-end imitation learning for HRI in [J4], Apr. 2024 - Nov. 2024.

Ehsan Ranjbari, Istituto Italiano di Tecnologia, supervised his work on haptic gloves and teleoperation of humanoid robots, and his contribution to the Workfar Syntro project; results in [C9], Jan. 2023 - Aug. 2024.

B.Sc. and M.Sc. thesis and internship supervision

Jod Baatiya, KAUST, internship on explicit inverse kinematics of serial manipulators, June – August 2026.

Chahine Ghannoudi, B.Sc. thesis, École Polytechnique de Tunisie, co-supervised his thesis with Prof. Shinkyu Park. Thesis on RL capable simulator for underwater vehicles, defended July 2026.

Jangkwon Cho, research internship, KAUST, internship on teleoperation of serial manipulators, January – February 2026.

Saverio Taliani, M.Sc. thesis, Istituto Italiano di Tecnologia and Sapienza University of Rome. Online nonlinear MPC for multimodal locomotion; results in [C10], defended September 2024.

Technical skills

Open-source software: author of the first release of **walking-teleoperation**, the teleoperation suite now maintained by the **robotology** organisation and used for whole-body teleoperation of the iCub and ergoCub humanoids; and of **stable-centroidal-mpc**, the reference implementation of the adaptive centroidal MPC of [J5] for bipedal (CasADi/IPOPT) and quadrupedal (ACADOS) platforms. Contributor to **YARP**, the **bipedal-locomotion-framework** and **RoboStack** (ros-humble).

Languages and tools: C/C++, MATLAB/Simulink, Python; CasADi, IPOPT, OSQP; CMake, Git, CI/CD, Pixi, Linux; PX4 Autopilot, PlatformIO. YARP, ROS and ROS 2, Gazebo, RViz.

Industrial automation: IEC 61131-3, Siemens TIA Portal and PCS7, Yokogawa Centum VP, Wonderware InTouch; P&ID and control architecture design, functional specifications, tendering.

Platforms: iCub and ergoCub humanoids, Unitree Aliengo quadruped, Crazyflie aerial vehicles, BlueBoat USV and custom towed underwater vehicle, VR and haptic peripherals.

Languages

Arabic native; English fluent; Italian basic.

References

Available upon request.